Title 3:

Prediction of Accuracy in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Random Forest Algorithm

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Keywords: Prediction, Accuracy, Extreme Gradient Boosting, Comparing, Random Forest, Inflation Rate.

Abstract:

Aim: This study aims to evaluate the effectiveness of different machine learning algorithms in raising the forecast accuracy of inflation rates. In order to ascertain which model produces the most accurate predictions, it specifically compares the performance of the XGBoost (extreme gradient booster) algorithm to that of the Random Forest method. **Materials And Methods:** A dataset of 15 years of previous rate of inflation data from 19 nations was used in the study. Key indicators of the economy, financial measurements, and policy changes influencing inflation patterns are all included in the dataset. The same data set and parameter settings were used to train both models, and each approach was evaluated ten times to ensure consistency. The recorded true-positive score was 0.132. **Result:** The findings show that the two approaches' prediction accuracy differs significantly. XGBoost outperformed the Random Forest method, whose had an accuracy percentage of 78.08%, with an expected accuracy of 90.11%. **Conclusion:** According to the study's findings, XGBoost outperforms the algorithm known as Random Forest in terms of inflation rate predicting accuracy. The most accurate model for predicting inflation is XGBoost, which has a prediction accuracy of 90.11% compared to 78.08%. For this reason, it is suggested as the best approach for financial analysts as well as managers who need accurate inflation rate projections.

Keywords: Accuracy, Comparison, Gradient Boosting, Extreme Gradient Boosting, Random Forest, Prediction, Inflation Rate

Introduction:

XGBoost, or extreme gradient boosting, has become well-known for its accuracy and efficiency in financial forecasting applications when contrasted with conventional machine learning techniques. For companies, investors, and politicians to make wise economic decisions, it is essential to forecast inflation rates with high accuracy. Interest rates, investment plans, and general economic stability are all impacted by inflation patterns, therefore using sophisticated machine learning models is crucial for accurate predictions. In order to ascertain whether approach offers better accuracy and resilience, this study evaluates how well XGBoost predicts inflation rates and contrasts its results with those of the Random Forest algorithm(“Global

Inflation Forecasting and Uncertainty Assessment: Comparing ARIMA with Advanced Machine Learning" 2025).

Analyzing historical inflationary data and assessing both algorithms' predictive power are the main goals of this study. XGBoost effectively manages big datasets and recognizes intricate links in economic indicators because to its gradient boosting structure. On another hand, a random forest, a method of ensemble learning based on trees of decisions, has grown widely employed for its capacity to minimize variation and increase generalization. Although each algorithm is superior in a different way, little is known about how well they compare directly in inflation forecasting. Finding the model with the best accuracy can greatly improve the dependability of economic forecasts(Huang, Qi, and Xia 2024).

The dynamic character of economic factors, which can change as a result of several regional and global circumstances, is one of the main obstacles to accurately forecasting inflation. A predictive model's capacity to adjust to these alterations and retain accuracy over time determines how successful it is. By comparing the advantages and disadvantages of XGBoost with Random Forest in terms of inflation rate predictions, this study seeks to close the gap in previous research. In order to improve decision-making in financial planning and economic policy, this study uses real-world data to determine the most dependable algorithm(Ray et al. 2025).

Materials and Methods:

The purpose of this study was to assess how well artificial intelligence models predicted inflation rates inside SIMATS. Two groups of models were created, one using the Random Forest method and the other the XGBoost technique. Each model was evaluated at various intervals, and statistical calculations were carried out utilizing alpha = 0.05 & beta = 0.2 to provide a fair comparison.

The dataset included important economic indicators, financial measures, and policy changes influencing inflation patterns, and it included 15 years' worth of inflation rate figures from several nations. Accurate analysis was ensured by applying standard pre-processing procedures to improve the quality of the data. More accurate predictions resulted from the application of a variety of data-cleaning techniques to preserve consistency and get rid of discrepancies.

Extreme Gradient Boosting:

Through an iterative boosting method that uses regularization techniques and weighted updates to reduce mistakes and increase prediction accuracy, XGBoost improves model performance. Among its benefits are:

1) Outstanding capacity to handle intricate connections
2) Efficiency of computation
3) Integrated regularization, which minimizes overfitting
4) Large dataset scalability
5) Choosing features wisely to improve model interpretability

These benefits have led to XGBoost's widespread recognition for its excellent accuracy, resilience, and scalability across a range of forecasting applications.

Random Forest:

To increase overall accuracy, the algorithm known as Random Forest builds many decision trees and aggregates their predictions. It is renowned for:

1) Excellent capacity for generalization
2) Effectively managing nonlinear interactions
3) Solidity against overfitting when properly adjusted
4) Consistency in various data distributions

This study investigates whether XGBoost outperforms Random Forest in inflation rate predictions, despite Random Forest being a potent predictive model.

Statistical Analysis:

The accuracy for the two forecasting methods was compared quantitatively using IBM SPSS. The models' accuracy differences were assessed for significance using cross-validation and hypothesis testing. The goal of the study was to determine the most effective strategy for improving the accuracy of inflation rate predictions.

Result:

With a precision of 90.11% as opposed to Random Forest's 73.24%, the comparison research showed that XGBoost fared better than Random Forest in forecasting inflation rates.

XGBoost consistently outperformed all test iterations, as seen in Table 3. Furthermore, Table 4's statistical analyses demonstrate the stability of XGBoost, with an average level of precision of 90.11% & a mean deviation (SD) at 0.084. Random Forest, on the other hand, had an average

success rate of 73.24%, having a standard deviation of 0.00527. These results demonstrate that XGBoost is the best technique for raising the accuracy of inflation rate predictions.

Discussion:

According to this study, the XGBoost (extreme gradient booster) method performs better than the Random Forest approach in the field of the rate of inflation forecasting, with a precision rate of 90.11% as opposed to 73.24%. The potential of XGBoost to better capture complex economic patterns was demonstrated by a paired t-test, which verified a significant distinction between the two models. The boosting method of XGBoost, which gradually enhances poor learners while lowering bias and variance, is primarily responsible for its exceptional performance. These results show that XGBoost outperforms Random Forest in terms of precision and stability when it comes to forecasting inflation trends(Vilenko 2025).

Because of its resilience and capacity to manage huge datasets without overfitting, Random Forest is still a formidable competitor. In contrast to boosting techniques like XGBoost, its emphasis on averaging tree decisions can occasionally result in less accurate predictions. XGBoost's optimization strategies, including regularization, parallel computing, and effective tree pruning, offer it an edge over its competitors in financial forecasting, even if Random Forest is superior in accessibility and noise reduction. This study backs up the claim that XGBoost is more accurate in forecasting inflation rates, especially in dynamic economic settings where precision is essential(Tsuchiya, Takemura, and Udagawa 2025).

These results are consistent with other studies that highlighted the benefits of boosting techniques in forecasting. The investigation emphasizes how crucial it is to choose models that strike a balance between precision, computing effectiveness, and flexibility. For greater prediction authority, subsequent investigations ought to explore the effects of more economically adjusted parameters, and combination techniques that combine the two approaches. Economists & finance professionals may enhance projections for inflation and make better policy and finance decisions by consistently improving forecasting models(Beer, Ferstl, and Graf 2025).

Conclusion:

According to the study's findings, XGBoost predicts inflation rates better than the Random Forest algorithm. Random Forest's accuracy was 78.08%, whereas XGBoost's precision was 90.11%. These findings suggest that XGBoost provides a more accurate and dependable way to predict inflation trends.

Declaration:

Conflicts of interest are absent from this paper.

Authors Contribution:

Author JSC was in charge of writing the manuscript as well as managing data collection and analysis. The text's critical evaluation, data validation, and conceptualization were all handled by author KVK.

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Table 1: pseudocode for XGBoost algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the XGBoost algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name XGBoost. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage.

Table 2: pseudocode for random forest algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the random forest algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name random forest. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage.

Table 3: Accuracy values of 10 iterations:

Iteration number	xg-boost accuracy	Rf accuracy
1	90.07%	73.24%
2	90.21%	73.24%
3	90.21%	73.24%
4	90.15%	73.25%
5	90.01%	73.25%
6	90.02%	73.25%
7	90.01%	73.25%
8	90.16%	73.25%
9	90.21%	73.24%
10	90.10%	73.24%

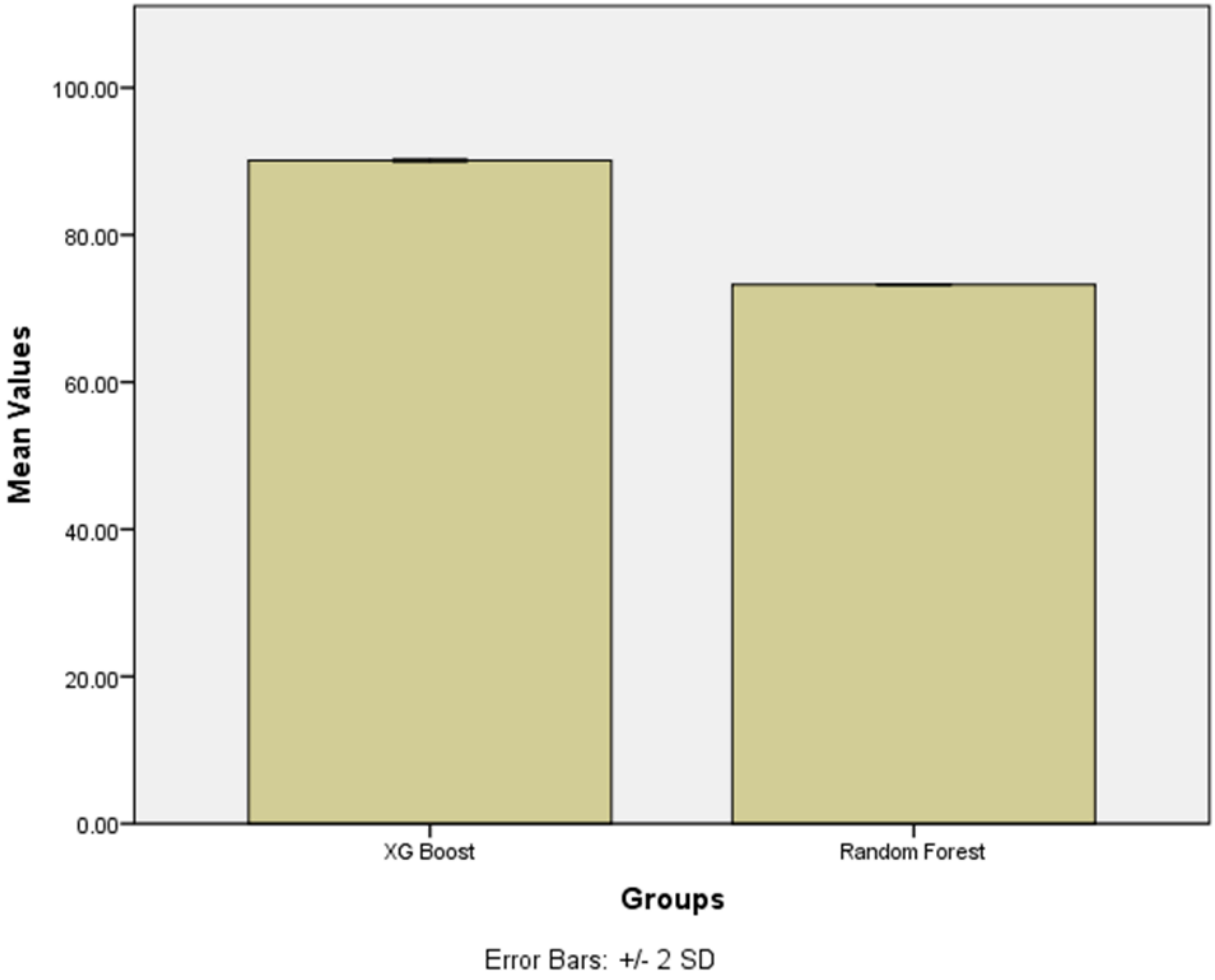
Table 4:

group	n	average	standard deviation	standard error mean
XGBoost	10	90.1150	0.08410	0.02659
rf	10	73.2450	0.00527	0.0167

Table 5: Independent Sample Test

independent samples test											
		levene's test for equality of variances		t-test for equality of means							
		f	sig.	t	df	sig. (2-tailed)	mean difference	std. error difference	95% confidence interval of the difference		
									lower	upper	
accuracy	equal variances assumed	0.52	0.121	451.478	18	0.00	12.03	0.02665	11.97	12.08	
	equal variances not assumed			451.478	9.071	0.00	12.03	0.02665	11.96	12.09	

Figure 1: Extreme Gradient Boosting algorithm **Vs** Gradient boosting algorithm



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